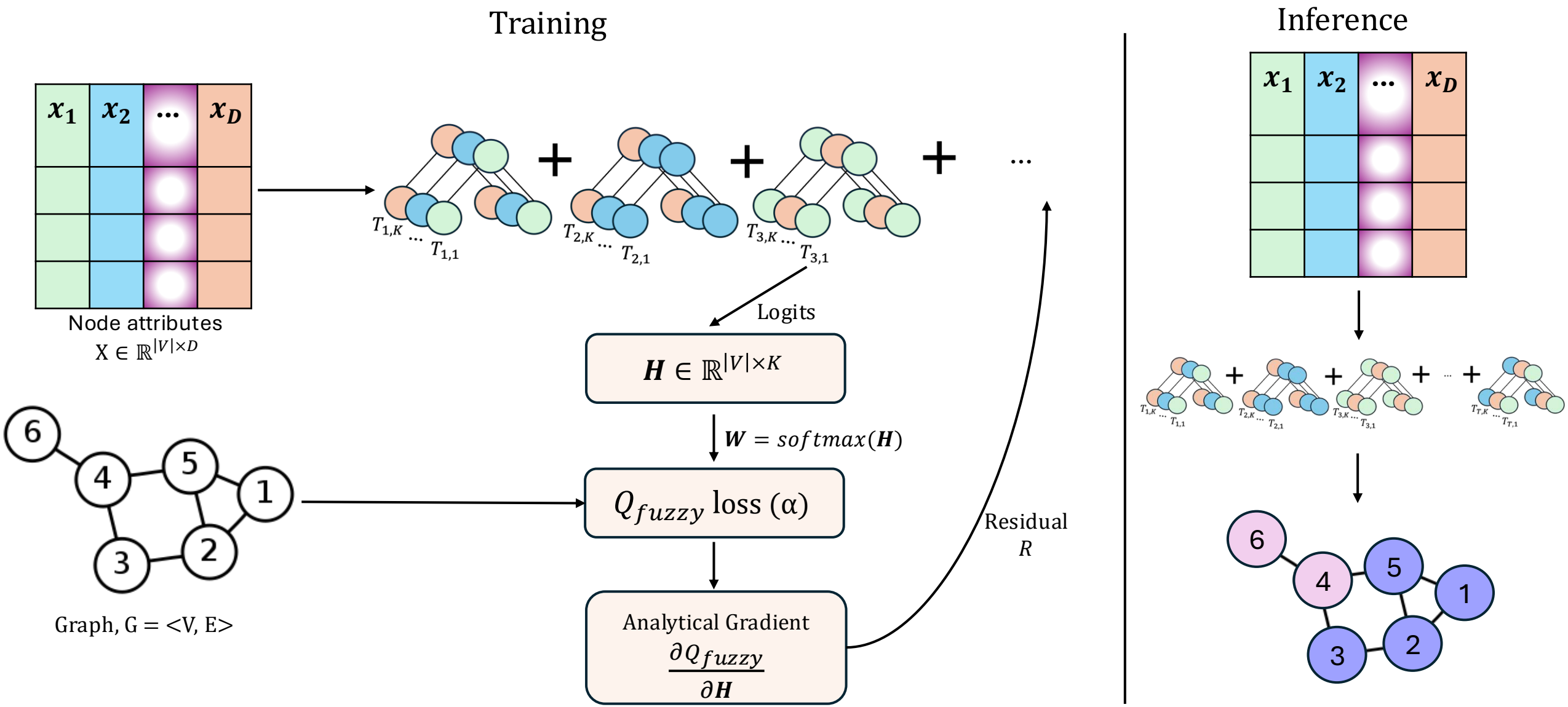Boosted-FSM-GAM